

Intergenerational Mobility Across U.S. Counties: Econometric Baselines, Machine Learning, and the Structure of Place-Based Opportunity

Econ 148 Project 3 — Track A

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Abstract

This paper studies why some U.S. counties generate more upward economic mobility than others. Using county-level mobility outcomes from Opportunity Insights and related datasets on segregation, school quality, family structure, inequality, commuting, social capital, poverty, income, population density, and urbanicity, we construct a county-level analytic sample of 1,382 counties. The outcome is absolute upward mobility: the mean adult income rank of children whose parents were at the 25th percentile of the national income distribution. The empirical design compares a transparent econometric baseline—OLS with standardized predictors and heteroskedasticity-robust standard errors—against a random forest trained on the same predictor set. OLS explains a large share of variation across counties ($R^2 = 0.636$), with social capital, single-parent share, median household income, school quality, commute time, and income segregation emerging as the clearest linear predictors. The random forest improves predictive performance ($R^2 = 0.677$ under 5-fold cross-validation; OOB $R^2 = 0.686$), indicating that non-linearities and interactions matter, but not enough to overturn the central substantive conclusion: social capital and family structure dominate the predictive structure of mobility. SHAP values and partial dependence plots show that social capital is the most important feature, single-parent share has the strongest negative relationship, and school quality and racial segregation enter through threshold-like and interaction-heavy patterns. The policy implication is that place-based opportunity is not reducible to a single dimension such as poverty or urbanicity; counties combine social connectedness, family structure, school quality, segregation, and poverty in ways that are jointly predictive of mobility.

1 Introduction

Intergenerational mobility is a direct measure of whether a child’s economic prospects are determined by parental income or by the environment in which the child grows up. In the United States, opportunity is highly geographic: children born to low-income parents in some counties grow up to reach substantially higher positions in the national income distribution than observationally similar children elsewhere. This raises a policy question that is both empirical and practical: which county characteristics are most predictive of upward mobility, and does a flexible machine-learning model reveal mechanisms that a linear econometric model misses?

This project answers that question using public county-level datasets associated with Raj Chetty and coauthors’ mobility research (Chetty et al., 2014; Opportunity Insights, n.d.). The main outcome is `kfr_pooled_p25`, the mean adult income rank for children with parents at the 25th percentile of the national income distribution. The predictors are chosen to match the core channels emphasized

in the mobility literature: segregation, school quality, family structure, income inequality, commuting patterns, social capital, poverty, median household income, and population density (Chetty et al., 2014). The project is designed around the Track A requirement to pose a clear economic question, describe public data construction, estimate an econometric baseline, compare it with a machine-learning model, and discuss what worked and what did not.

The central empirical result is that the linear model is already strong: standardized OLS explains 63.6 percent of the variation in county-level upward mobility. That performance is substantively meaningful because the outcome is a noisy, long-run social object: adult income ranks reflect decades of family, neighborhood, school, labor-market, and institutional inputs. At the same time, the random forest improves predictive power to 67.7 percent under 5-fold cross-validation, with an out-of-bag score of 68.6 percent. The improvement is not huge, but it is large enough to suggest that non-linearities and interactions are real features of the mobility process rather than modeling noise.

The paper therefore does not treat machine learning as a black-box replacement for econometrics. Instead, the random forest is used as a diagnostic tool: if OLS captures the average linear relationships, SHAP feature importance, SHAP interaction values, and partial dependence plots help identify where the linear model is too rigid. This framing preserves the interpretability of OLS while using ML to test the substantive claim that social mobility depends on combinations of local conditions rather than on additive one-variable channels.

2 Data and Measurement

2.1 Data sources and merge design

The analytic dataset merges seven county-level public data sources on 5-digit FIPS codes. The mobility outcome and race-by-sex mobility measures come from the Opportunity Atlas county outcomes file (Opportunity Insights, n.d.). Older county-level mobility covariates are taken from *Where is the Land of Opportunity?* and related Geography of Mobility data (Chetty et al., 2014). The project also uses Opportunity Insights county covariates for commute time, poverty, median household income, and population density (Opportunity Insights, n.d.); Race and Economic Opportunity Table 8 for Gini and single-parent measures (Chetty et al., 2020); Health Inequality Project Table 12 for racial segregation, income segregation, income-adjusted school quality, and pupil-teacher ratio; Social Capital I data for economic connectedness (Chetty et al., 2022); and NCHS urban-rural codes for the six-category urban-rural classification.

The merge starts from the Opportunity Atlas county outcomes file and left-joins the covariate sources. This design keeps the universe of counties with mobility outcomes and then discards observations with missing values in the final modeling variables. After complete-case filtering, the analytic sample contains $N = 1,382$ counties. The dropped observations are not arbitrary: the final predictor expansion from nine to eleven predictors, especially the addition of pupil-teacher ratio, reduces the sample relative to earlier drafts. That sample-size tradeoff is justified because the expanded specification captures two paired constructs that matter substantively: racial and income segregation as distinct forms of spatial separation, and school quality as both an income-adjusted output measure and a resource-input measure.

For reproducibility, the submitted zip file includes the runnable analysis notebook, raw public data files, processed intermediate files, and generated figures. Raw data are stored in `data/raw/`, cleaned outputs in `data/processed/`, and report figures in `figures/`. The notebook is designed to run from start to

finish without manual intervention.

2.2 Outcome and predictors

The primary outcome is upward mobility,

$$\text{mobility}_c = \text{kfr_pooled_pooled_p25}_c,$$

where c indexes counties. It is interpreted as the mean adult income rank of children whose parents were at the 25th percentile of the national income distribution.

The predictor set contains eleven variables. The main structural variables are racial segregation (Theil index), income segregation (Theil index), income-adjusted school quality, pupil-teacher ratio, single-parent share, Gini inequality, mean commute time, and social capital measured by economic connectedness. The control variables are poverty share, median household income, and population density. All predictors are standardized before OLS and random-forest estimation so coefficient magnitudes and model splits are not driven by units of measurement.

Two measurement choices are especially important. First, earlier drafts used Black poverty share as a racial-economic concentration proxy, but diagnostic checks showed that it was essentially uncorrelated with proper Theil-index segregation measures. The final model therefore replaces that proxy with `cs_race_theil_2000` for racial segregation and adds `cs00_seg_inc` for income segregation. Second, the school-quality measure is changed from raw third-grade math scores to `score_r`, an income-adjusted test-score percentile that is closer to a value-added school-quality measure than to a level-of-affluence measure.

2.3 Temporal alignment and limitations in measurement

The data are intentionally year-mixed. The mobility outcome reflects children born around 1978–1983 and adult outcomes observed around 2014–2015 (Chetty et al., 2014; Opportunity Insights, n.d.). Segregation and commute variables are drawn from Census 2000; single-parent share, inequality, poverty, and population density are primarily 2010-era variables; school variables are approximately 2009–2013; median household income is 2016; and social capital is measured using Facebook friendship data from 2022 (Chetty et al., 2022). This is standard in the Chetty-style county mobility literature because the relevant county-level traits are highly persistent, but it is still a limitation. The largest temporal stretch is social capital: children in the outcome cohort were shaped by social connectedness in the 1980s and 1990s, while the observed measure comes from 2022. The interpretation therefore relies on the persistence of local connectedness over time.

3 Exploratory Evidence

The first step is to understand the distribution of upward mobility across counties and the degree to which broad geographic classes explain it. Figure 1 shows that mobility is centered around a median income rank of 0.397, with substantial but not extreme dispersion. The urban-rural boxplot shows meaningful overlap across all six NCHS classes. Large fringe metros appear somewhat higher on average, but no urban-rural category fully dominates the others. This suggests that urbanicity matters, but it is not the sole organizing variable for opportunity.

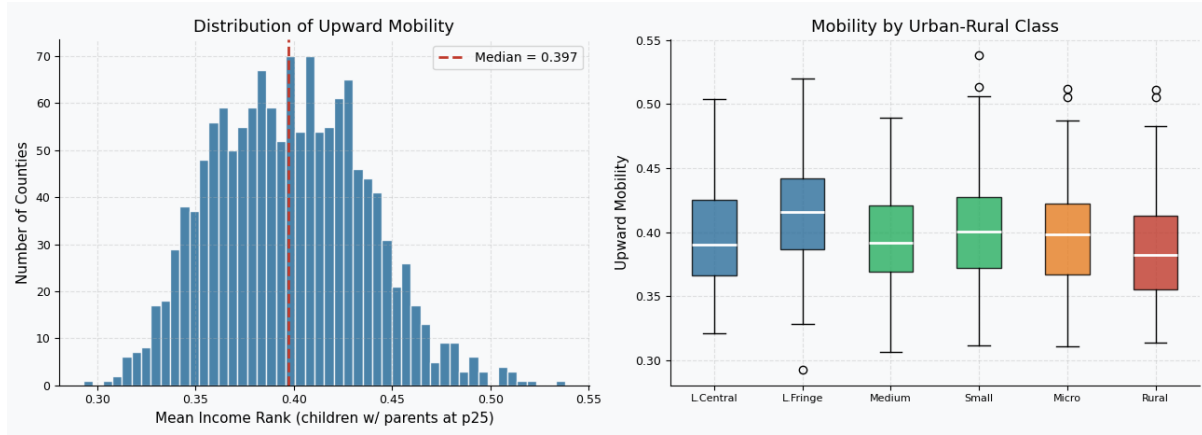


Figure 1: Distribution of upward mobility and mobility by urban-rural class. The outcome is the mean adult income rank for children whose parents were at the 25th percentile.

The bivariate evidence clarifies the dominant linear relationships. Figure 2 reports the correlation matrix among mobility and the main predictors. Social capital has the strongest positive correlation with mobility ($r = 0.75$), while single-parent share ($r = -0.68$) and poverty share ($r = -0.57$) have the strongest negative correlations. School quality is also positively correlated with mobility ($r = 0.46$). By contrast, income segregation and racial segregation have weaker unconditional correlations with mobility, even though they are conceptually important and correlated with other county characteristics. The off-diagonal cells also show the key multicollinearity problem: low-mobility counties tend to bundle high poverty, high single-parent share, low social capital, and lower school quality. OLS coefficients must therefore be interpreted as conditional relationships, not as isolated bivariate facts.

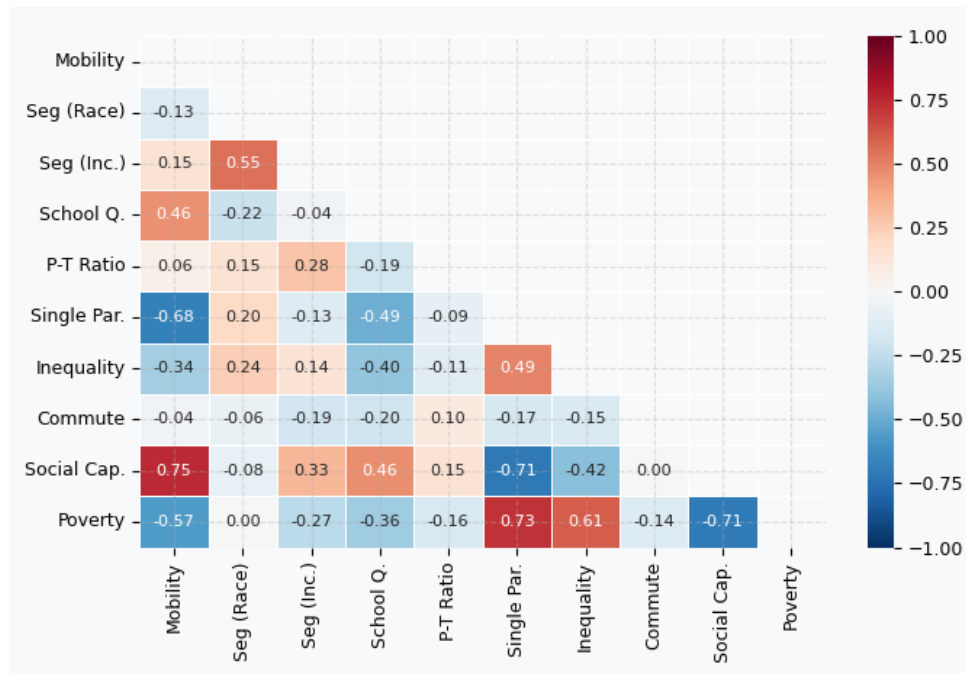


Figure 2: Correlation matrix of mobility and main predictors. Social capital and single-parent share are the strongest unconditional predictors, while poverty, school quality, and the segregation measures are embedded in a correlated county-characteristics block.

The segregation-poverty regime analysis provides a more structural view. Counties are split into four groups using above- and below-median racial segregation and poverty. Figure 3 shows that low-segregation/low-poverty and high-segregation/low-poverty counties have higher mobility distributions than the two high-poverty regimes. The high-segregation/high-poverty group performs poorly, but the visual comparison indicates that poverty is not simply interchangeable with segregation: the high-segregation/low-poverty group remains relatively high, while low-segregation/high-poverty counties perform weakly. The interaction between segregation and poverty is therefore not mechanically additive; the same level of segregation has different implications depending on the poverty environment.

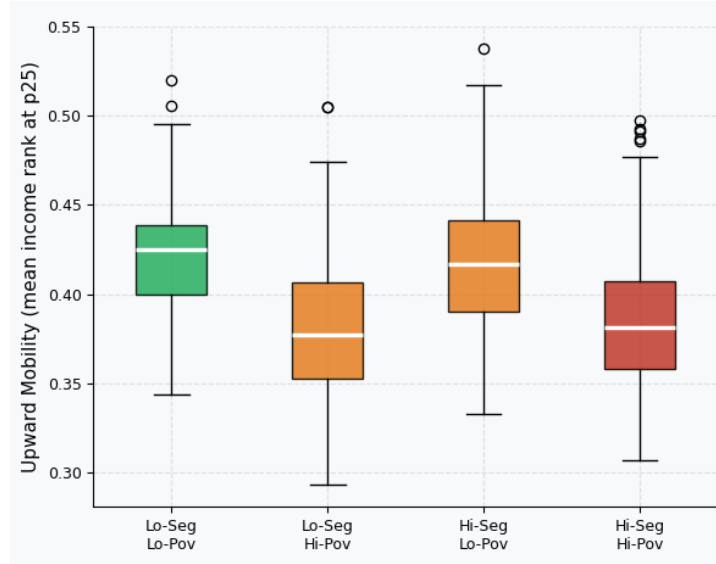


Figure 3: Mobility by racial-segregation and poverty regime. Counties are split by above/below-median racial segregation and above/below-median poverty.

Figure 4 examines a secondary outcome: the White-Black male mobility gap by urban-rural class. The key pattern is that the gap is almost always positive: White male mobility exceeds Black male mobility in nearly every county-class panel. Moreover, the slope of the gap with respect to overall mobility is generally positive, meaning higher overall mobility does not automatically close racial gaps. This matters for policy interpretation. A county can be high-opportunity on average while still producing unequal mobility gains across race and sex groups.

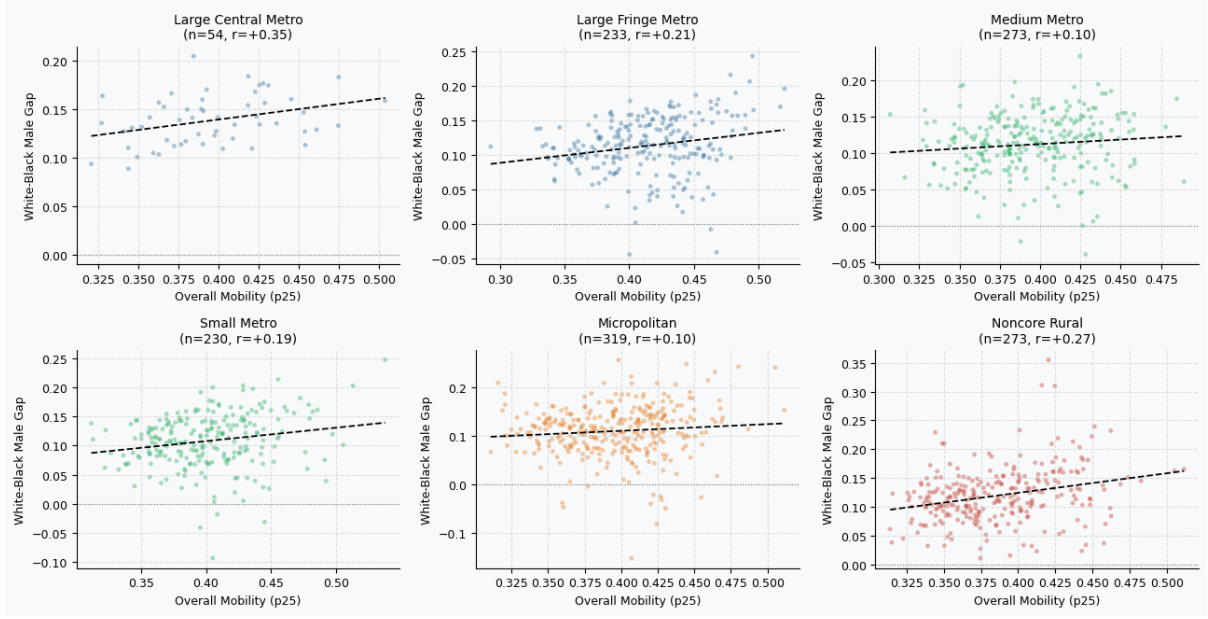


Figure 4: White-Black male mobility gap against overall mobility, separately by urban-rural class. Positive gaps dominate across the county distribution.

4 Econometric Baseline

4.1 Baseline specification

The baseline econometric model is an additive OLS regression of county mobility on the standardized predictor vector:

$$\text{mobility}_c = \alpha + X'_c \beta + \varepsilon_c,$$

where X_c contains racial segregation, income segregation, school quality, pupil-teacher ratio, single-parent share, Gini inequality, commute time, social capital, poverty share, median household income, and population density. All standard errors are HC3 heteroskedasticity-robust.

The interaction model extends the baseline with four theory-motivated cross terms:

$$\begin{aligned} \text{mobility}_c = & \alpha + X'_c \beta + \gamma_1(\text{seg_race}_c \times \text{school_quality}_c) \\ & + \gamma_2(\text{seg_race}_c \times \text{single_parent}_c) \\ & + \gamma_3(\text{inequality}_c \times \text{single_parent}_c) \\ & + \gamma_4(\text{seg_race}_c \times \text{social_capital}_c) + \varepsilon_c. \end{aligned}$$

The interaction set is deliberately small and theory-driven: it tests whether racial segregation modifies the relationship between mobility and schools, family structure, and social capital, and whether inequality interacts with family structure.

4.2 OLS results

Figure 5 shows the standardized OLS coefficients for the baseline and interaction models. The baseline model reaches $R^2 = 0.636$, while adding the four interaction terms raises R^2 only to 0.639. The linear relationships are therefore stable: the interaction model barely improves fit.

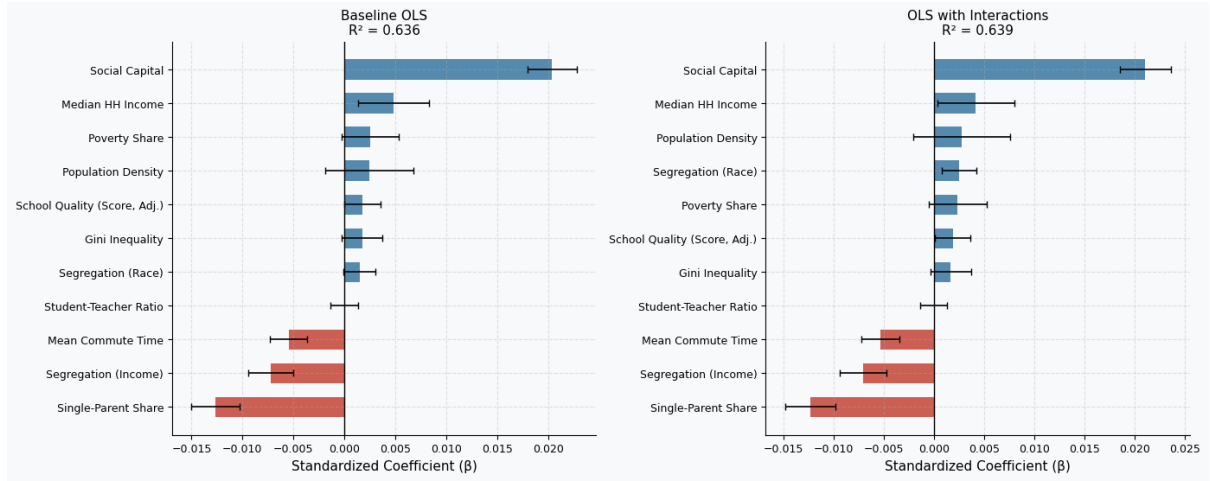


Figure 5: OLS estimates for the baseline model and the interaction model. Predictors are standardized; error bars represent robust confidence intervals.

The coefficient pattern is intuitive and consistent with the exploratory analysis. Social capital is the largest positive predictor. Single-parent share is the largest negative predictor. Median household income is positive, while mean commute time and income segregation are negative. School quality is positive but smaller than social capital. Population density, student-teacher ratio, and some inequality terms have weaker conditional effects.

Among the four interaction terms, only racial segregation $\times$ single-parent share crosses the conventional 5 percent significance threshold ($\beta \approx -0.002$, $p \approx 0.017$). The racial segregation $\times$ school quality, inequality $\times$ single-parent share, and racial segregation $\times$ social capital interactions are statistically null. This is an important modeling result: OLS can detect interactions when they are specified correctly, but it does not provide a general search mechanism for discovering interactions. The machine-learning section below addresses exactly that limitation.

4.3 Urban-rural heterogeneity

The same baseline regression is also estimated within each of the six NCHS urban-rural classes, with predictors re-standardized within class. Figure 6 shows that the broad signs are fairly robust: social capital is positive across every class, and single-parent share is negative across every class. However, the magnitude of these relationships varies substantially. For example, the negative association of single-parent share appears especially large in large central metros, and the confidence intervals widen in smaller strata. This supports a heterogeneous-place interpretation: the same predictor set matters everywhere, but the intensity of each mechanism differs across the urban-rural hierarchy.

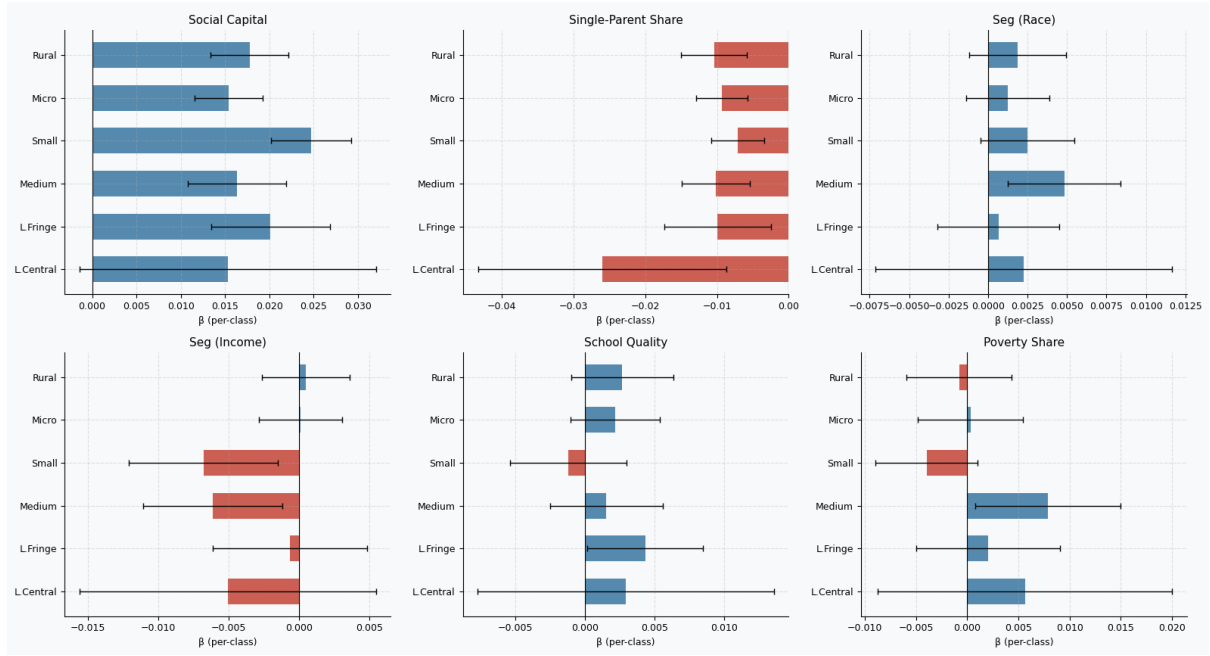


Figure 6: OLS coefficients by urban-rural class for selected predictors. Predictors are re-standardized within each class, so coefficients should be read as within-class standardized relationships.

5 Machine-Learning Comparison

5.1 Random forest design

The random forest is trained on the same eleven standardized predictors as the OLS model. The model uses 500 trees, `max_features=0.33`, `min_samples_leaf=5`, out-of-bag scoring, and 5-fold cross-validation. These hyperparameters are intentionally conservative. With only 1,382 counties, an aggressive grid search could overfit the validation folds themselves. The goal is not to maximize a leaderboard score but to test whether a flexible model finds predictive structure that OLS misses.

Figure 7 summarizes the performance comparison. OLS explains 63.6 percent of variation, while the random forest reaches 68.6 percent out-of-bag and 67.7 percent under 5-fold cross-validation. Because the reported random-forest comparison uses out-of-sample scoring rather than in-sample fit, the improvement is evidence of real non-linear structure rather than pure overfitting.

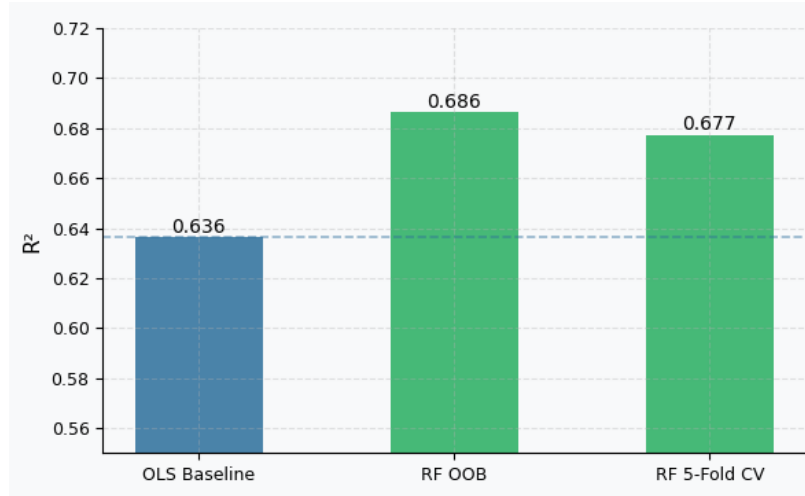


Figure 7: Model R^2 comparison. The random forest improves on OLS under both out-of-bag scoring and 5-fold cross-validation.

5.2 Feature importance and interaction structure

SHAP values are used to interpret the fitted random forest. Figure 8 shows that social capital is the dominant feature, followed by single-parent share, school quality, median household income, population density, poverty share, and commute time. The ordering is consistent with OLS for the top variables, but SHAP also reveals that some variables matter more through thresholds and interactions than through smooth linear slopes.

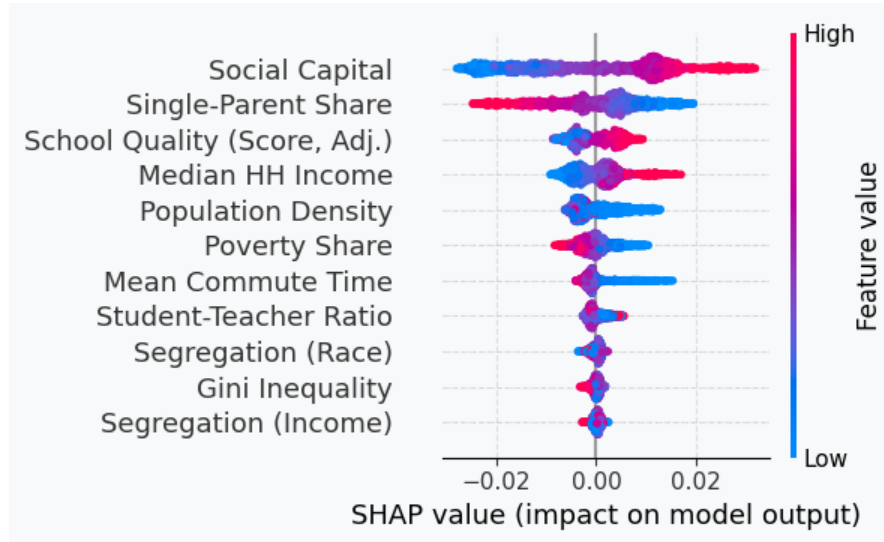


Figure 8: SHAP feature importance from the random forest. Social capital and single-parent share dominate the predictive structure, while school quality and income variables contribute secondary but meaningful information.

Table 1 combines the OLS and SHAP summaries. The table shows that social capital has the largest OLS coefficient and the largest SHAP importance. Single-parent share is second by SHAP importance and strongly negative in OLS. School quality is statistically significant but smaller in OLS, while SHAP gives it a larger role, suggesting that school quality enters the random forest in a more non-linear way. Median household income is significant in OLS but has lower SHAP importance than social capital and

family structure. Segregation measures are weaker as main effects but are more visible in interaction-based diagnostics.

OLS R ² : 0.636 RF CV R ² : 0.677 RF OOB R ² : 0.686 N = 1382						
Out[]:	Predictor	OLS β	Sig.	OLS p-value	SHAP Importance	SHAP Interaction Sum
0	Social Capital	0.0204	***	0.000	0.0130	0.0051
1	Single-Parent Share	-0.0127	***	0.000	0.0074	0.0036
2	School Quality (Score, Adj.)	0.0018	**	0.048	0.0038	0.0025
3	Median HH Income	0.0048	***	0.007	0.0036	0.0027
4	Population Density	0.0025		0.267	0.0035	0.0018
5	Poverty Share	0.0026	*	0.072	0.0024	0.0022
6	Mean Commute Time	-0.0055	***	0.000	0.0020	0.0013
7	Student-Teacher Ratio	0.0000		0.986	0.0013	0.0011
8	Segregation (Race)	0.0015	*	0.060	0.0007	0.0008
9	Segregation (Income)	-0.0072	***	0.000	0.0005	0.0009
10	Gini Inequality	0.0018	*	0.087	0.0005	0.0008

Table 1: Final combined results table: standardized OLS coefficients, OLS significance, SHAP feature importance, and SHAP interaction sums.

Figure 9 reports SHAP interaction values. The strongest interactions involve social capital, single-parent share, poverty, median household income, and school quality. This confirms the main modeling interpretation: the random forest improves on OLS not because it discovers an entirely different set of variables, but because it captures how the key variables condition each other. Low social capital is especially costly in counties that also have high single-parent rates or poverty, and school quality interacts with the broader social environment rather than operating as an isolated input.

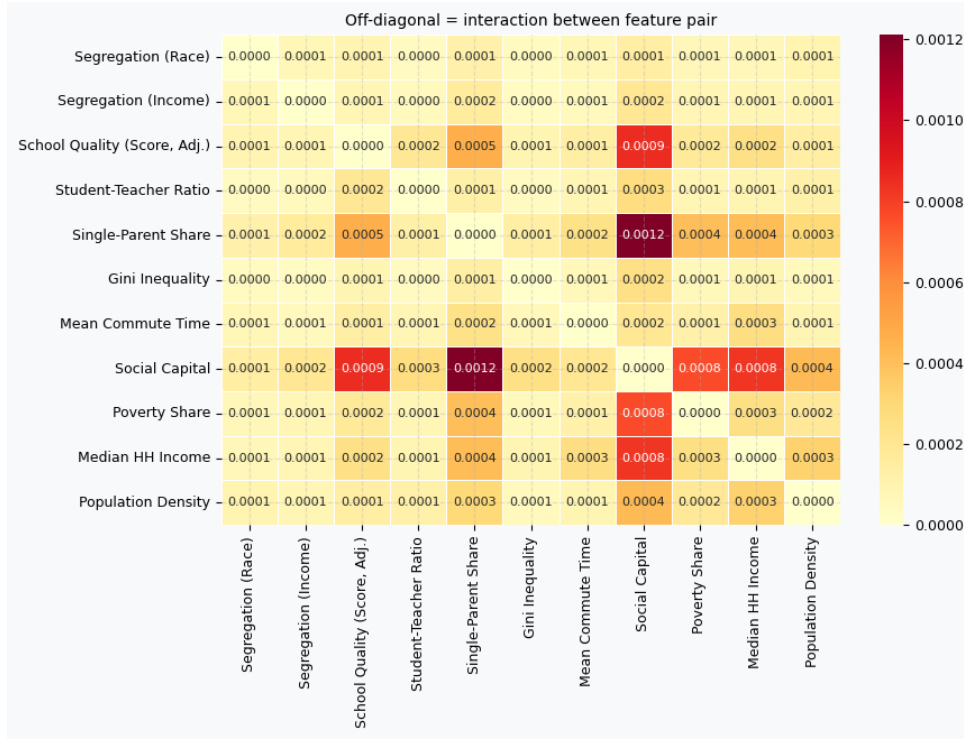


Figure 9: SHAP interaction values, measured as mean absolute interaction effects. The strongest interaction structure is concentrated among social capital, single-parent share, poverty, median household income, and school quality.

Partial dependence plots make the non-linear patterns more interpretable. Figure 10 shows monotonic and threshold-like relationships for the four most important features. Social capital has a strong increasing relationship with predicted mobility, with especially large gains over the middle of its distribution. Single-parent share has a clear decreasing relationship. School quality is weak at low values but rises sharply around the middle-to-high range, suggesting threshold effects. Racial segregation is non-monotonic: the fitted relationship changes direction across the distribution, which helps explain why its linear OLS coefficient is relatively weak even though segregation appears in interaction diagnostics.

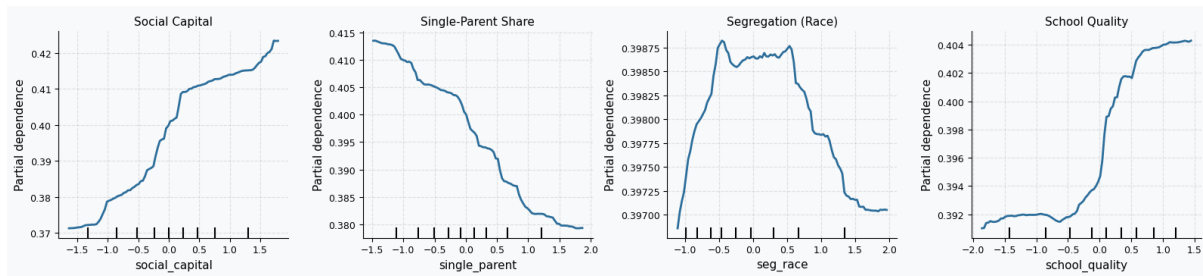


Figure 10: Partial dependence plots for the top four random-forest features. Social capital and single-parent share show strong directional relationships, while school quality and racial segregation exhibit more non-linear patterns.

6 Discussion

6.1 What worked

The project works best as a combination of econometric transparency and ML-based diagnosis. OLS provides a clean baseline: standardized coefficients identify the average conditional relationship between each county characteristic and upward mobility. The baseline is strong enough that the main empirical story is not fragile. The strongest predictors are stable across exploratory correlations, OLS coefficients, and SHAP values: social capital is strongly positive, single-parent share is strongly negative, and school quality, poverty, commute time, and median household income matter as secondary predictors.

The random forest adds value by identifying where the additive linear model is incomplete. Its cross-validated improvement over OLS is modest but real. That pattern is exactly what one should expect if the true mobility process is mostly structured by a few strong county characteristics but also shaped by thresholds and interactions. The partial dependence plots show that school quality and segregation are not well summarized by one slope. The SHAP interaction matrix shows that the social-capital/family-structure/poverty block contains meaningful interaction structure. In substantive terms, mobility is a system outcome: the same level of school quality or segregation can mean different things depending on the surrounding social and economic structure.

The regime and race-gap analyses also work as interpretive complements. They show that aggregate mobility and equity are not the same outcome. Higher overall mobility does not guarantee smaller White-Black male gaps, and high-segregation/high-poverty regimes are not merely low on the average mobility distribution; they also reveal demographic inequality patterns that are masked by the pooled outcome.

6.2 What did not work as cleanly

The largest limitation is not model performance but identification. The models are predictive and descriptive, not causal. A positive social-capital coefficient does not prove that increasing economic connectedness mechanically raises mobility by that coefficient amount. Social capital is correlated with many unobserved county traits: school institutions, labor-market access, civic capacity, local governance, housing patterns, and historical racial sorting. OLS controls for observable covariates, and the random forest captures non-linear observables, but neither design solves omitted-variable bias.

Second, the year-mixing creates measurement risk. Social capital is measured in 2022 even though the relevant childhood exposure window for the outcome cohort is decades earlier. The analysis relies on persistence in county traits. That assumption is plausible for segregation, inequality, and many measures of social connectedness, but it is still a substantive assumption rather than a directly tested fact. Commute time also comes from 2000, while several other predictors are 2010-era variables. These timing mismatches likely add measurement error and may attenuate or distort coefficients.

Third, complete-case filtering reduces the sample from the full county universe to 1,382 counties. This improves internal consistency by ensuring every model uses the same observations, but it can reduce external validity if missingness is systematic. The pupil-teacher ratio variable is a major driver of the sample reduction. If counties missing this variable differ systematically in rurality, size, or administrative reporting quality, the final analytic sample may overrepresent counties with better data availability.

Fourth, the random forest improves prediction but does not automatically translate into causal policy prescriptions. SHAP values and partial dependence plots are model-interpretation tools. They describe how the trained model uses features, not what would happen under an intervention. The policy lesson

should therefore be framed cautiously: the most predictive counties combine high social capital, lower single-parent share, stronger school quality, and lower poverty, but the estimates do not identify the causal return to changing any one of those variables in isolation.

6.3 Policy interpretation

The main policy implication is that mobility-enhancing interventions are likely complementary. The OLS model says social capital and family structure are the dominant average predictors. The random forest says that those predictors interact with poverty, school quality, income, and segregation. A county with weak social capital and high poverty is not simply the sum of two disadvantages; the combination itself is predictive of lower mobility. Likewise, school quality appears to matter more through thresholds and interactions than through a smooth linear effect.

This pushes against one-dimensional place policy. A narrow intervention focused only on poverty reduction, only on school quality, or only on commuting access may miss the fact that opportunity is jointly produced by the local social environment, family structure, and institutions. The strongest evidence points toward integrated policies: strengthening cross-class social connectedness, supporting stable family and neighborhood environments, improving school quality, and reducing concentrated poverty and segregation at the same time. The ML comparison does not replace the economic model; it reinforces the economic intuition that mobility is produced by systems of county characteristics rather than by isolated variables.

7 AI Usage Disclosure

AI assistance was used during the data-analysis workflow, not as a substitute for the group’s economic framing or final analytical judgment. The tool used was Claude Code (Anthropic), Opus 4.7, run inside VS Code as an interactive coding assistant. AI assistance supported data engineering, feature engineering, modeling setup, interpretation tooling, plotting, documentation, and code quality. Specific tasks included drafting the multi-source FIPS merge, constructing engineered variables and regime labels, setting up OLS models with HC3 standard errors, implementing the random forest with 5-fold cross-validation, generating SHAP summaries and partial dependence plots, defining the matplotlib visual style, drafting notebook markdown, and refactoring plotting helper functions to satisfy modularity requirements.

AI assistance was also used in two substantive debugging and refinement steps. First, it helped investigate whether Health Inequality Project Table 12 merged cleanly with the analytic sample and whether income-adjusted `score_r`, income segregation, and pupil-teacher ratio improved the predictor set. Second, it helped diagnose that the earlier `seg_score` variable was better interpreted as Black poverty share rather than a true segregation measure. The final specification therefore replaced that variable with a proper racial-segregation Theil index and added income segregation as a separate predictor.

Verification was performed through end-to-end notebook execution, sample-size accounting after the complete-case drop, cross-checks of coefficient directions against the Chetty literature, out-of-sample random-forest scoring, and a code-style audit. The scope of authorship remains the group’s: the research question, economic framing, predictor choices, interpretation of results, and final write-up decisions were made by the group. AI assistance accelerated implementation and documentation but did not determine the substantive argument.

8 Conclusion

This paper uses county-level public data to model upward mobility for children from low-income families. The econometric baseline and the machine-learning comparison produce a consistent result: county opportunity is most strongly associated with social capital and family structure, with school quality, poverty, commute time, income, and segregation adding important secondary and interaction-based information. OLS explains 63.6 percent of the variation in upward mobility, while a random forest improves cross-validated performance to 67.7 percent. The improvement indicates that non-linearities and interactions matter, but the ML model does not overturn the linear story; it sharpens it.

The strongest interpretation is that place-based mobility is produced by bundles of county characteristics. High social capital, lower single-parent share, better school quality, lower poverty, and less harmful forms of segregation tend to appear together, and the random forest shows that their combinations matter. At the same time, the analysis is descriptive rather than causal, and its strongest limitations are temporal mismatch, missingness, and unobserved confounding. The results should therefore be read as a structured predictive map of mobility-relevant county characteristics, not as a causal estimate of any single intervention. Still, as a policy guide, the evidence is clear: improving mobility requires attention to the social and institutional ecosystem of counties, not only to one isolated economic variable.

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Group Contributions

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